

ADVAITH RENJITH

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EDUCATION

Rutgers University — New Brunswick, NJ <i>B.S. Mechanical Engineering</i>	<i>August 2026 – May 2030</i>
Edison Academy Magnet School — Edison, NJ	<i>Class of 2026</i>

STANDARDIZED TEST SCORES

SAT Superscore	1580 / 1600	Math: 800	Reading & Writing: 780
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AP EXAMINATIONS — 12 EXAMS TOTAL

Score of 5 (8 exams) AP Physics C: Mechanics · AP Physics C: E&M · AP Calculus BC · AP Statistics · AP Physics 1 · AP Literature · AP Psychology · APUSH
Score of 4 (4 exams) AP Biology · AP Chemistry · AP English Language · AP Microeconomics · AP Macroeconomics

COMPETITIONS & AWARDS

Sir Isaac Newton Physics Exam <i>9th Place Internationally — 3,800 global participants</i>	<i>May 2025</i>
USA Physics Olympiad (USAPhO) <i>Distinction — Top 400 of 6,000 national participants</i>	<i>April 2025</i>
F=MA Examination — USAPhO Qualifier	<i>February 2025</i>
NASA App Development Challenge <i>National Winner — 200+ competing teams</i> Built 3D visualization of the Artemis II lunar mission in Unreal Engine, requiring orbital mechanics modeling and advanced 3D graphics programming	<i>2025</i>
MIT Beaver Works Summer Institute <i>4th Place Nationally — 100 competing teams</i> Led CubeSat design project for satellite-based wildfire detection, achieving theoretical 90% fire detection accuracy	<i>2025</i>
NJ State Science Fair — CS Division <i>2nd Place — Physics-Informed Neural Network Research</i>	<i>2026</i>

RESEARCH & PUBLICATIONS

NLLP 2025 Workshop @ EMNLP — Third Author

Accepted 2025

"Beyond the Haystack: Sensitivity to Context in Legal Reference Recall"

- Perfect reviews from all reviewers; introduced evaluation framework measuring LLM recall from U.S. court opinion briefings
- Exposed limitations of standard long-context benchmarks such as Needle in a Haystack across three naturalistic legal recall tasks

Physics-Informed Neural Network Research

Senior Year, 2025–2026

- Designed hyperlocalized multimodal Physics-Informed Diffusion Graph Transformer with on-device quantum layer; 2nd place NJ State Science Fair

Multi-Tiered Drone Swarm Research

Junior Year, 2024–2025

- Developed RL models for autonomous multi-agent wildfire detection combining robotics, computer vision, and environmental modeling

TECHNICAL PROJECTS

Mixture of Agents Deep Research System

2025–2026

- On-device system using centralized reasoner and specialist sub-models; matched performance of cloud-hosted monolithic systems
- Integrated browser use, computer use, fact-checking, and verified output generation entirely on-device

Soft Robotics Arm

2024–2026

- 3-cable TPU-segment arm using differential cable tension and material flexibility for full 3D spatial movement
- Developing VLM + heuristic solver control approach alongside pose-driven actuation methods

Active Aerodynamics Testbed (RC Car)

2023–2026

- RC car with active downforce fan, DRS, on-device AI perception, and ECU running on repurposed smartphone

VOLUNTEERING & LEADERSHIP — 700+ CONFIRMED HOURS

NJ Aviation Hall of Fame

2024–2026

- Digitizing 150+ years of aviation history across 8,000+ pages of scrapbooks into a secure private high-resolution archive

InfoAge Science & History Museums

2024–2026

- Docent, repairs technician, and social media manager across radio observatory, vintage computer collection, and makerspace
- Building unified multi-museum artifact database spanning 500+ years of history across 20 buildings

STEM Tutor & Mentor / Nonprofit Technology Consultant

Ongoing

- Tutoring advanced mathematics, physics, and CS to younger students
- Pro-bono web development for multiple nonprofits; 700+ confirmed volunteer hours